



TikTok to collect biometric data

TikTok secretly updated their privacy policy to collect users' biometric data. <https://digit.in/jul21-41>



World's thinnest technology

Researchers have engineered the world's thinnest technology, with a thickness of only two atoms. <https://digit.in/jul21-42>

dVerse is taking the licensing route, they responded, “yes.... We are in talks with elevator manufacturers to help us reach the technology to more people, and are already in talks with major elevator companies like Kone, Johnson and Schindler.” When asked about the next steps for the startup, dVerse tells us, “We are looking at tackling the problem of hospital-acquired infections from a system perspective by first tackling individual nodes within the hospital. We are an impact-focused company. With the rising oxygen crisis in the second wave, we are developing oxygen concentrators based on the open-source project, Oxikit at an affordable price using locally sourced components.” Then, dVerse told us how Nexus helped the startup, “Nexus was a great learning experience and we feel much more confident as entrepreneurs. We had focused sessions on finance, marketing and great workshops from industry experts. Nexus creates a safe space for everyone to voice their opinions and doubts without the fear of being judged. It is definitely the best place for tech entrepreneurs to gain more practical knowledge about how to go about running a business.”

DRONA AUTOMATIONS IS REPLACING MANUAL SCAVENGERS WITH ROBOTS

Manual scavenging is a health hazard, and an unsafe occupation that has a lot of associated problems.

Still, drains have to be unclogged and cleaned, and here robots are particularly suitable for the task. Drona Automations is a robotics company that is developing robots for a wide range of applications, from sewage to space. Beyond being functional, these robots are designed to have a social impact as well, reducing the exploitation of cheap labour. For the cleaning robots, the challenges are making robots small enough to easily fit into the pipes, and at the same time prevent any damage to the plumbing. Drona Automations is not afraid of taking a crack at some really hard problems, “Drona Automations is aimed at ending manual scavenging of manholes by using low cost and easy to use robots. The company is also developing robots which can be used to clean the sewer pipes of manholes and also any kind of sewer pipes. In the current market, cleaning and inspection of sewer pipes is not easy. By using our robots, we can reduce the downtime of cleaning and clean & inspect pipes of any size from 3 inches to 12 inches without any damage to the pipes.”

Drona Automations then took us through the process of developing a custom robot for the particular application, such as the swage cleaning robot. “First, we conduct detailed survey and collect data and suggestions to all the stakeholders involved. Then we perform a 3D model of the entire robot depending on the level of customization required. The model will be in accordance with latest DFMA methods. After that, we manufacture a prototype and field test it. Then, after a few design enhancements, we will finalize the product

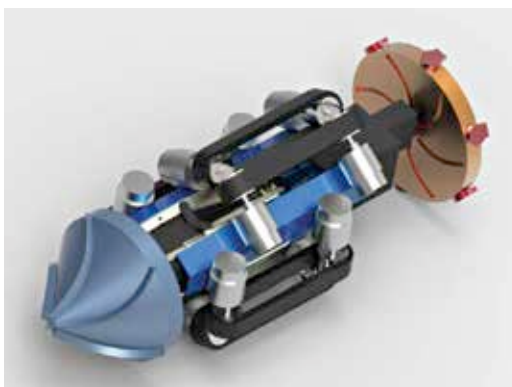


Drona Automations Horizontal Sewer Pipe Cleaning Robot (HSPCR)

design.” We then asked about the response to the sewage robots and the societal impact, Drona Automations told us, “using an automated machine or a system to clean sewer pipes has seen a lot of attraction from the end users as it reduces the manual scavenging of manholes and for the ease of cleaning. In India, manual scavenging of manholes is still happening in even metropolitan cities. This is mainly because there is no efficient system in place to clean the sewer pipes. All the systems will be used for cleaning the vertical manholes instead of the pipes. Using our robots, we can use it to clean and inspect the pipes. Technically, the automated cleaning robots are what can be called drones, and the company has plans of using its expertise to clean up not just the drains, but the low earth orbit as well. Drona Automations is also planning on developing automated fleets of satellites that can be used to clear up space junk, “drones are essentially any automated / semi automated machine which can be used to perform any desired action. Drones can be used to clear our space debris because we are in a era of Space Pollution.” The Nexus accelerator helped Drona Automations with developing their logo, and two products, the Manhole Sewer Pipe Cleaning Robot (MSPCR) and the Horizontal Sewer Pipe Cleaning Robot (HSPCR).

DELIVERAIN

Deliverain is an augmented reality platform that provides tech to support maintenance, repair and overhaul tasks for industrial workers.



Drona Automations Manhole Sewer Pipe Cleaning Robot (MSPCR)